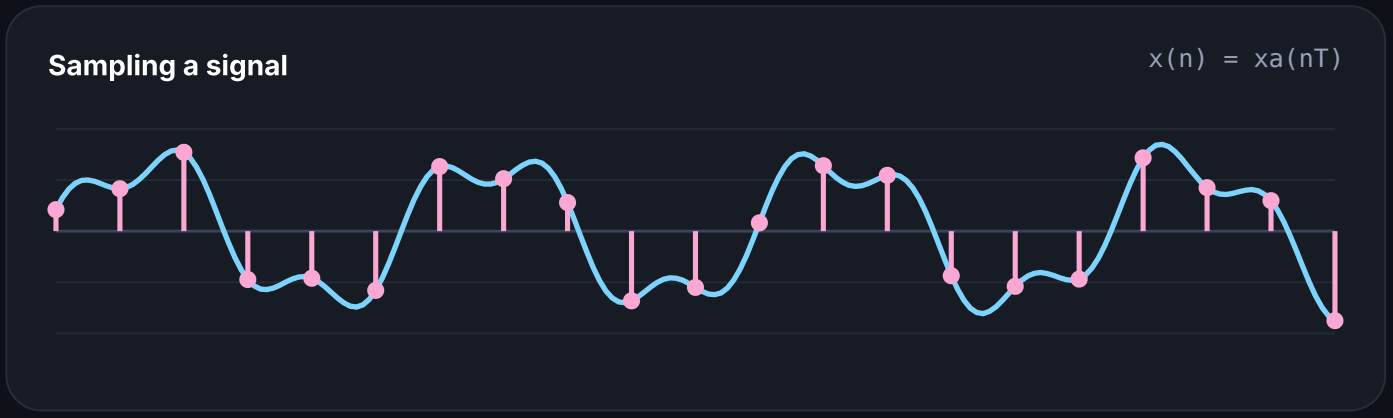
المسائل و Sheet 1
Chapter 1

Digital Signal Processing · Problems & Sheet 1 · 24 questions



Signals & Systems | Frequency | A/D & D/A | Sampling | Aliasing | Quantization

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Lecturer: Dr. Samir M. Kamal

Modern Academy for Engineering & Technology

DSP / ch01 / contents

المحتوى

Problems & Sheet 1 · 24 questions

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إزاي تذاكر من الملف

المسائل خطوة بخطوة بنفس طريقة الدكتور: أمثلة المحاضرات، الامتحانات القديمة، والشيت والأسايمنت. تحت كل مسألة مربع النتيجة، وفي آخر كل جزء صفحة بكل النتايج.

SECTION 01

مسائل المحاضرات

Examples

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×3 P1

Example 1.1 — Sampling & aliasing

Consider the analog signal $x_a(t) = 3 \cos 100\pi t$.

a) Determine the minimum sampling rate required to avoid aliasing.

b) The signal is sampled at Fs = 200 Hz. What is the discrete-time signal?

c) The signal is sampled at Fs = 75 Hz. What is the discrete-time signal?

d) What is the frequency $0 < f < Fs/2$ of a sinusoid that yields samples identical to those in (c)?

a) The frequency of the analog signal is f = 50 Hz. Hence the minimum sampling rate required to avoid aliasing is Fs = 100 Hz.

b) If the signal is sampled at Fs = 200 Hz, the discrete-time signal is

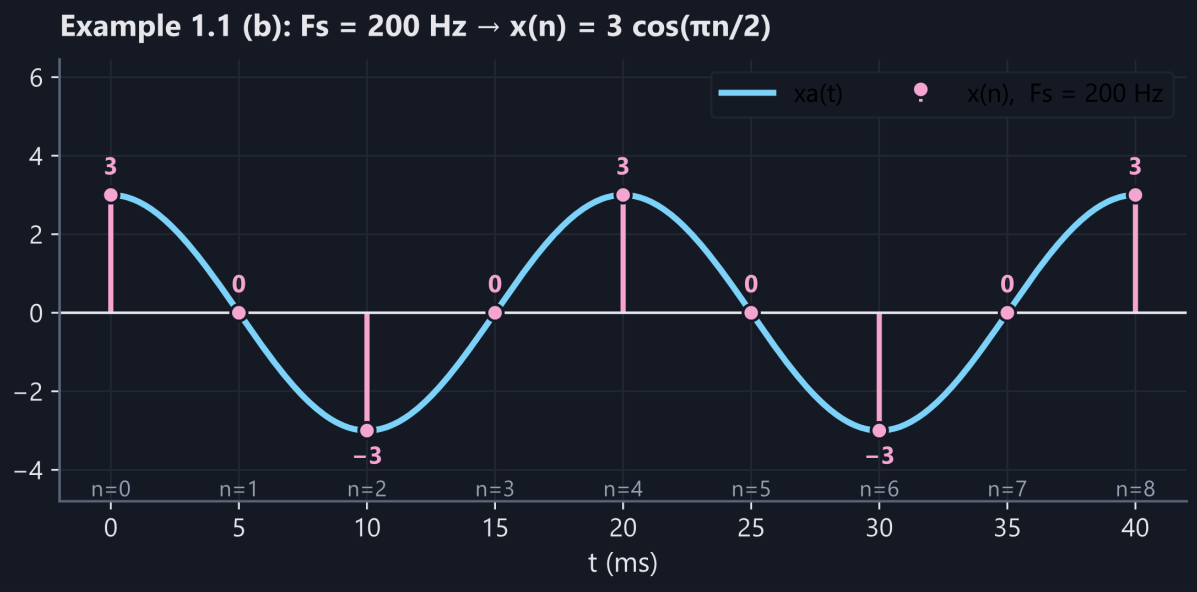
$x(n) = 3 \cos(100\pi/200)n = 3 \cos(\pi/2)n$

c) If the signal is sampled at Fs = 75 Hz, the discrete-time signal is

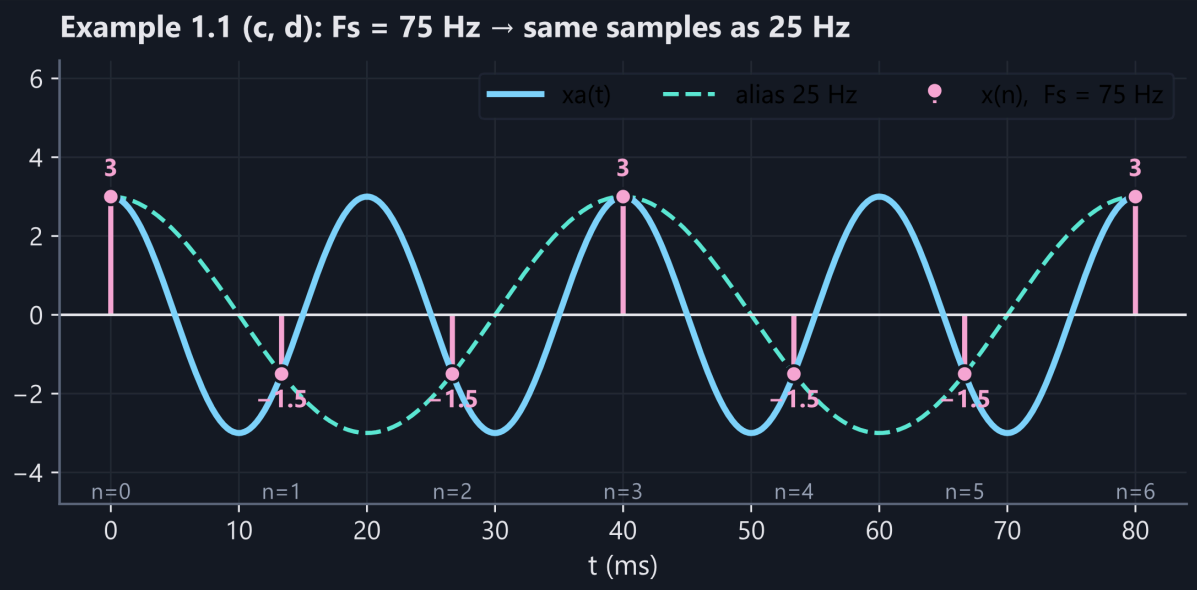
$x(n) = 3 \cos(100\pi/75)n = 3 \cos(4\pi/3)n = 3 \cos(2\pi - 2\pi/3)n = 3 \cos(2\pi/3)n$

d) For the sampling rate of Fs = 75 Hz, we have f = F·Fs = 75F. The frequency of the sinusoid in part (c) is F = 1/3. Hence f = 25 Hz.

Clearly, the sinusoidal signal ya(t) = 3 cos 2πft = 3 cos 50πt sampled at Fs = 75 samples/s yields identical samples. Hence f = 50 Hz is an alias of f = 25 Hz for the sampling rate Fs = 75 Hz.



مسائل المحاضرات



النتيجة

$F_{s,min} = 100$ Hz · $F_s = 200$ Hz → $x(n) = 3\cos(\pi n/2)$ · $F_s = 75$ Hz → $x(n) = 3\cos(2\pi n/3)$ · 25 Hz gives the same samples (50 Hz is its alias)

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 4 · Example 1.1 · p. 38

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 4- Ch- 1.pdf

جه في

Final 20-1-2025 | Final 11-6-2023 | Final 23-1-2023 (نفس النوع: Fs كافي ولا لأ + x(n))

5-Exams/Final/

×3 P2

Example 1.2 — Nyquist rate

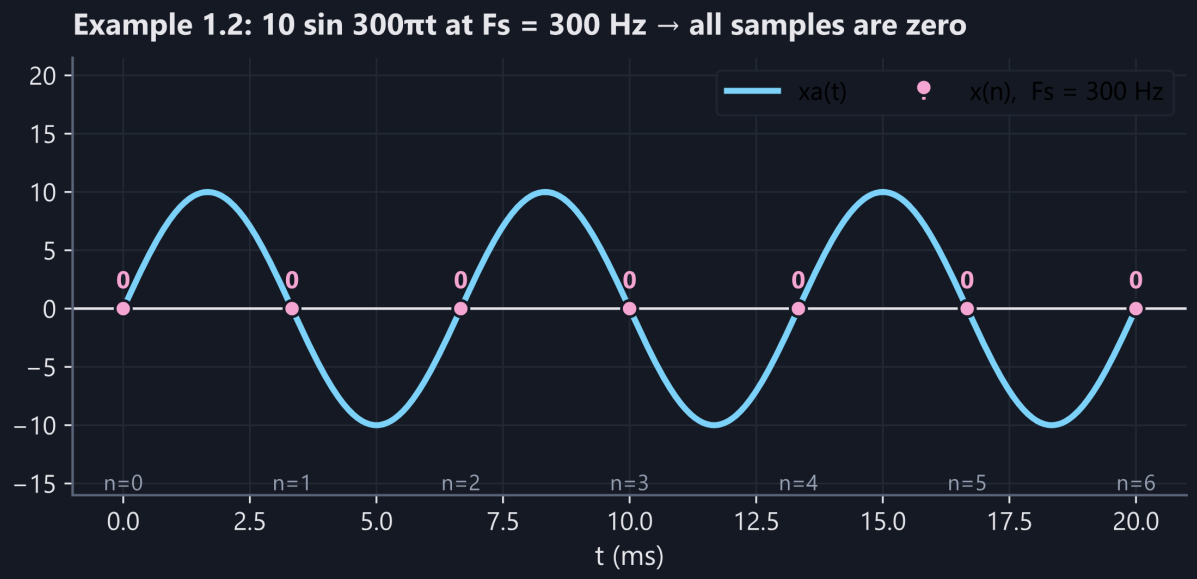
Consider $x_a(t) = 3 \cos 50\pi t + 10 \sin 300\pi t - \cos 100\pi t$. What is the Nyquist rate for this signal?

The frequencies present in the signal above are f1 = 25 Hz, f2 = 150 Hz, f3 = 50 Hz. Thus fmax = 150 Hz and according to (1.47), Fs > 2fmax = 300 Hz. The Nyquist rate is fN = 2fmax. Hence fN = 300 Hz.

Discussion

- It should be observed that the signal component 10 sin 300πt, sampled at the Nyquist rate fN = 300, results in the samples 10 sin πn, which are identically zero.
- In other words, we are sampling the analog sinusoid at its zero-crossing points, and hence we miss this signal component completely.
- This situation would not occur if the sinusoid is offset in phase by some amount θ. In such a case we have 10 sin(300πt + θ) sampled at the Nyquist rate fN = 300 samples per second, which yields the samples:
  $10 \sin(\pi n + \theta) = 10(\sin \pi n \cos \theta + \cos \pi n \sin \theta) = 10 \sin \theta \cos \pi n = (-1)^n 10 \sin \theta$
- Thus if $\theta \neq 0$ or π, the samples of the sinusoid taken at the Nyquist rate are not all zero.
- However, we still cannot obtain the correct amplitude from the samples when the phase θ is unknown.
- A simple remedy that avoids this potentially troublesome situation is to sample the analog signal at a rate higher than the Nyquist rate.

مسائل المحاضرات



النتيجة

fN = 300 Hz; sampling exactly at fN can miss the 150 Hz component (samples = $(-1)^n$ 10 sin θ) → sample above the Nyquist rate

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 4 · Example 1.2 · p. 41

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5-Exams/Final/

P3

Example 1.3 — Aliasing with three components

Consider $x_a(t) = 3\cos 2000\pi t + 5\sin 6000\pi t + 10\cos 12000\pi t$.

a) What is the Nyquist rate?

b) The signal is sampled at Fs = 5000 samples/s. What is the discrete-time signal?

c) What is the analog signal $y_a(t)$ reconstructed using ideal interpolation?

(a) The frequencies existing in the analog signal are f1 = 1 kHz, f2 = 3 kHz, f3 = 6 kHz. Thus fmax = 6 kHz, and according to the sampling theorem, Fs > 2fmax = 12 kHz. The Nyquist rate is fN = 12 kHz.

(b) Since we have chosen Fs = 5 kHz, the folding frequency is Fs/2 = 2.5 kHz, and this is the maximum frequency that can be represented uniquely by the sampled signal. By making use of (1.30) we obtain:

$x(n) = xa(nT) = xa(n/Fs)$
$= 3\cos 2\pi(1/5)n + 5\sin 2\pi(3/5)n + 10\cos 2\pi(6/5)n$
$= 3\cos 2\pi(1/5)n + 5\sin 2\pi(1 - 2/5)n + 10\cos 2\pi(1 + 1/5)n$
$= 3\cos 2\pi(1/5)n + 5\sin 2\pi(-2/5)n + 10\cos 2\pi(1/5)n$

Finally, we obtain:

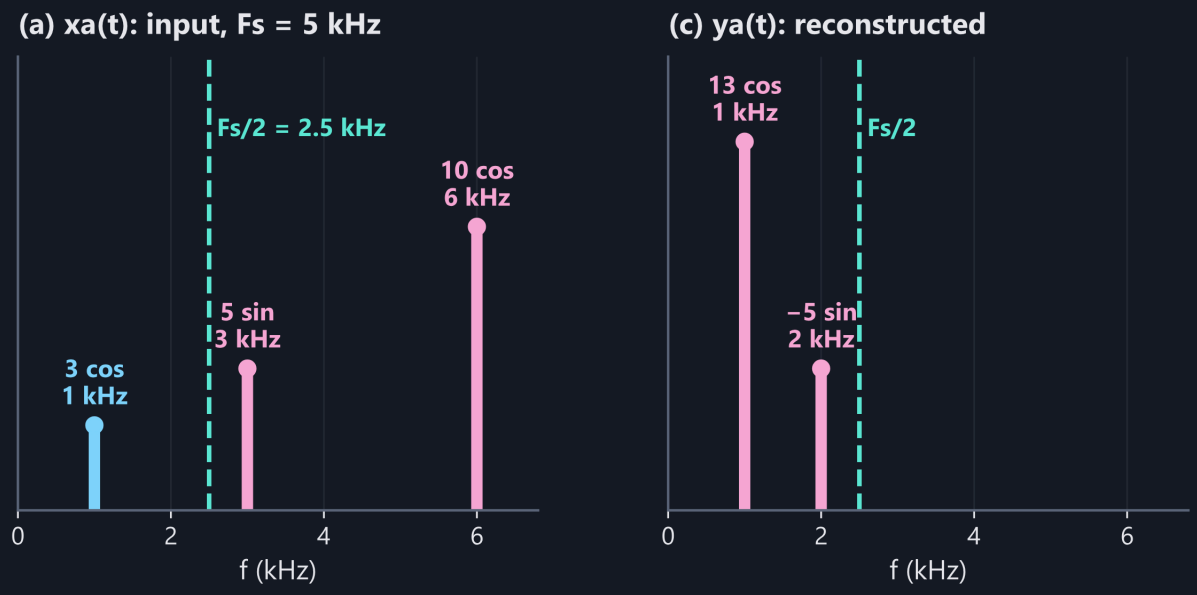
$x(n) = 13\cos 2\pi(1/5)n - 5\sin 2\pi(2/5)n$

(c) Since only the frequency components at 1 kHz and 2 kHz are present in the sampled signal, the analog signal we can recover is

$ya(t) = 13\cos 2000\pi t - 5\sin 4000\pi t$

which is obviously different from the original signal xa(t). This distortion of the original analog signal was caused by the aliasing effect, due to the low sampling rate used.

مسائل المحاضرات



النتيجة

$f_N = 12$ kHz · $x(n) = 13\cos 2\pi(1/5)n - 5\sin 2\pi(2/5)n$ · $y_a(t) = 13\cos 2000\pi t - 5\sin 4000\pi t$ (aliasing)

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 4 · Example 1.3 · p. 44

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P4

Example 1.4 — The aliasing concept

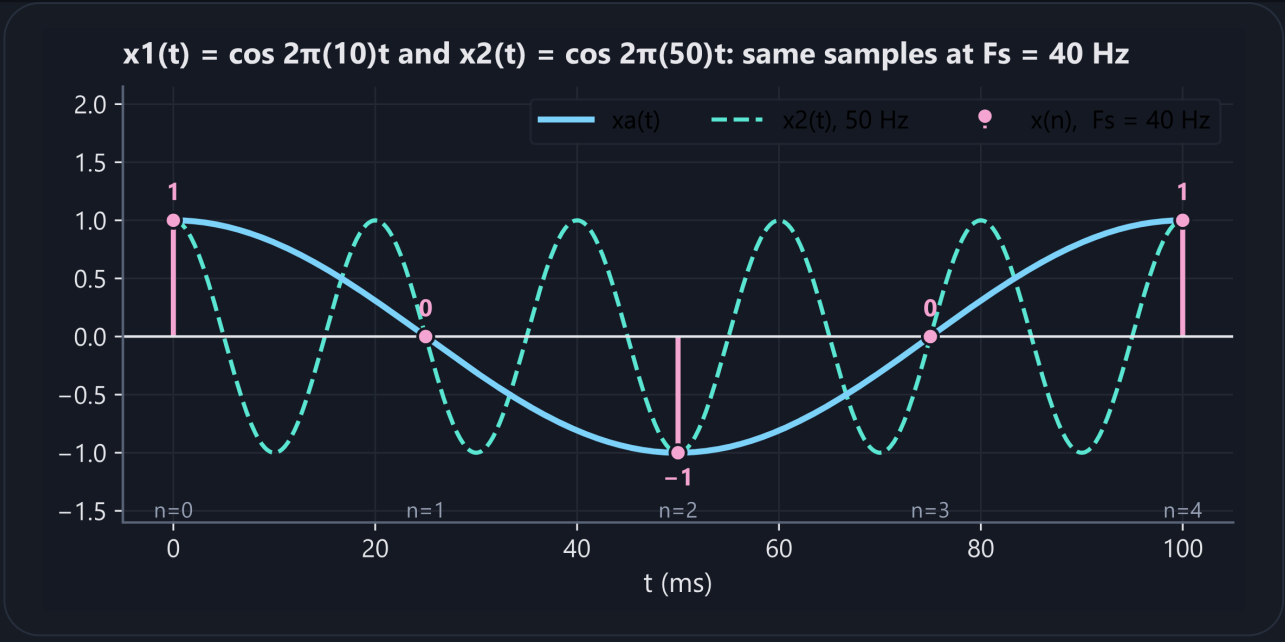
Two signals x1(t) = cos 2π(10)t and x2(t) = cos 2π(50)t are sampled at Fs = 40 Hz. Find x1(n) and x2(n) and discuss.

The implications of these frequency relations can be fully appreciated by considering the two analog sinusoidal signals x1(t) = cos 2π(10)t and x2(t) = cos 2π(50)t, which are sampled at a rate Fs = 40 Hz. The corresponding discrete-time signals or sequences are:

x1(n) = cos 2π(10/40)n = cos (π/2)n
x2(n) = cos 2π(50/40)n = cos (5π/2)n

- However, cos 5πn/2 = cos(2πn + πn/2) = cos πn/2. Hence x2(n) = x1(n).
- Thus the sinusoidal signals are identical and, consequently, indistinguishable.
- If we are given the sampled values generated by cos(π/2)n, there is some ambiguity as to whether these sampled values correspond to x1(t) or x2(t).
- Since x2(t) yields exactly the same values as x1(t) when the two are sampled at Fs = 40 samples per second, we say that the frequency f2 = 50 Hz is an alias of the frequency f1 = 10 Hz at the sampling rate of 40 samples per second.
- It is important to note that f2 is not the only alias of f1. In fact at the sampling rate of 40 samples per second, the frequency f3 = 90 Hz is also an alias of f1, as is the frequency f4 = 130 Hz, and so on.

مسائل المحاضرات



النتيجة At Fs = 40 Hz, f2 = 50 Hz is an alias of f1 = 10 Hz (so are 90 Hz, 130 Hz, …)

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 4** · Example 1.4 · p. 48

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 4- Ch- 1.pdf

P5

Quantization example (Table 1.2)

$x_a(t) = 0.9^t$, $t \geq 0$ is sampled at Fs = 1 Hz, giving $x(n) = 0.9^n$. Quantize the first 10 samples to one significant digit by truncation and by rounding; find the error, Δ, and the number of bits.

- Let us consider the discrete-time signal $x(n) = 0.9^n$ for $n \geq 0$ (0 for $n < 0$), obtained by sampling the analog exponential signal $xa(t) = 0.9^t$, $t \geq 0$ with a sampling frequency Fs = 1 Hz (Ts = 1 sec), see Fig. 1.15(a).
- Observation of Table 1.2, which shows the values of the first 10 samples of x(n), reveals that the description of the sample value x(n) requires n significant digits.
- However, let us assume that we want to use only one significant digit. To eliminate the excess digits, we can either simply discard them (truncation) or discard them by rounding the resulting number (rounding).
- The resulting quantized signals xq(n) are shown in Table 1.2. We discuss only quantization by rounding, although it is just as easy to treat truncation. The rounding process is graphically illustrated in Fig. 1.15b.
- In our example we have xmax = 1, xmin = 0, and L = 11, which leads to Δ = 0.1.
- In our example it can easily be seen that we need a coder with b = 4 bits.

n	x(n)	Truncation	Rounding	eq(n)
0	1	1.0	1.0	0.0
1	0.9	0.9	0.9	0.0
2	0.81	0.8	0.8	−0.01
3	0.729	0.7	0.7	−0.029
4	0.6561	0.6	0.7	0.0439
5	0.59049	0.5	0.6	0.00951
6	0.531441	0.5	0.5	−0.031441
7	0.4782969	0.4	0.5	0.0217031
8	0.43046721	0.4	0.4	−0.03046721

مسائل المحاضرات

9	0.387420489	0.3	0.4	0.012579511

$$\Delta = (x_{max} - x_{min})/(L - 1) = (1 - 0)/(11 - 1) = \mathbf{0.1} \qquad e_q(n) = x_q(n) - x(n),$$
$$|e_q(n)| \le \Delta/2 = 0.05$$
$$2^b \ge L = 11 \rightarrow \mathbf{b = 4\ bits}$$



النتيجة $\Delta = 0.1,\ L = 11,\ |eq(n)| \le \Delta/2 = 0.05,\ b = 4$ bits

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 4 · Table 1.2 · p. 27

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 4- Ch- 1.pdf

P6

Periodicity examples (Lec 3)

a) Find the fundamental period of a DT sinusoid with F1 = 31/60 and F2 = 30/60.

b) For x(n) = cos $\Omega_o n$, find N for Ω_o = 0, π/8, π/4, π/2, π.

c) Fig. 1.7: Ω = π/6. Find F and N.

F = k/N (1.16): cancel common factors so that k and N are relatively prime → N is the fundamental period

a) F1 = 31/60 → k = 31, N = 60 (relatively prime) → **N1 = 60**. F2 = 30/60 = 1/2 → **N2 = 2**. A small change in frequency results in a large change in the period.

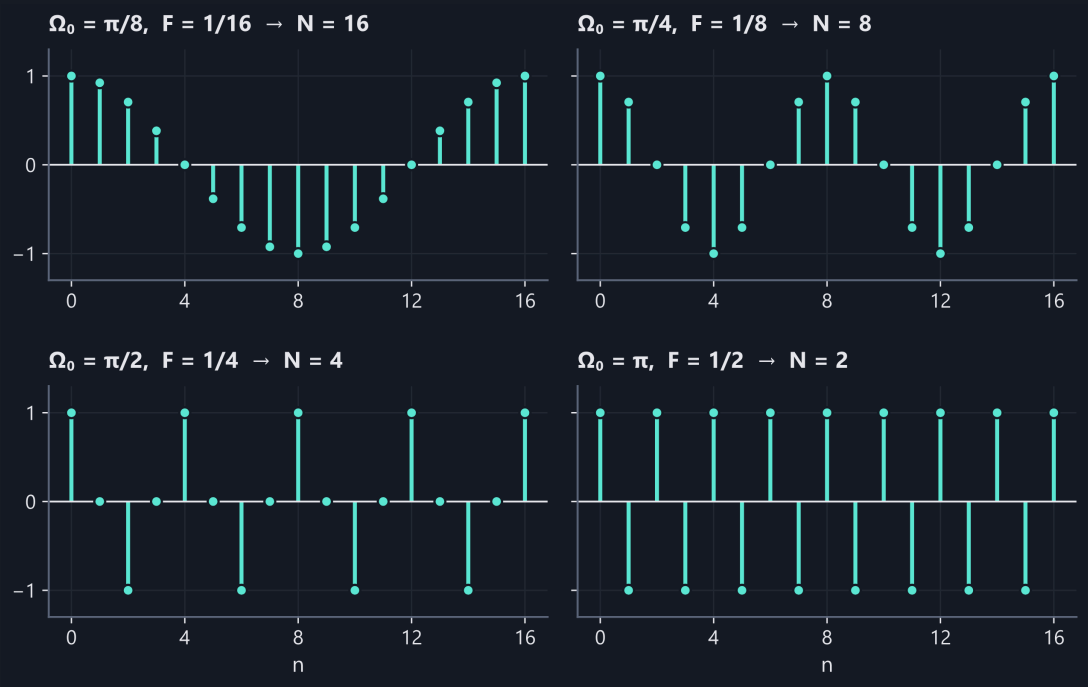
b) x(n) = cos $\Omega_0 n$, F = $\Omega_0/2\pi$:

$\Omega_0 = 0 \rightarrow F = 0 \rightarrow N = \infty$ | $\pi/8 \rightarrow 1/16 \rightarrow N = 16$ | $\pi/4 \rightarrow 1/8 \rightarrow N = 8$

$\pi/2 \rightarrow 1/4 \rightarrow N = 4$ | $\pi \rightarrow 1/2 \rightarrow N = 2$

The period decreases as the frequency increases (highest rate of oscillation at Ω = π).

c) Ω = π/6 → F = (π/6)/2π = 1/12 → **N = 12**



مسائل المحاضرات

النتيجة a) N1 = 60, N2 = 2 · b) Ω₀ = 0, π/8, π/4, π/2, π → N = ∞, 16, 8, 4, 2 · c) N = 12

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 3 · Sec. 1.5.2 · p. 9
1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 3- Ch- 1.pdf

★ Cheat sheet · مسائل المحاضرات

النتايج النهائية

Fs,min = 100 Hz · Fs = 200 Hz → x(n) = 3 cos(πn/2) · Fs = 75 Hz → x(n) = 3 cos(2πn/3) · 25 Hz gives the same samples (50 Hz is its alias)	P1
fN = 300 Hz; sampling exactly at fN can miss the 150 Hz component (samples = (−1)ⁿ 10 sin θ) → sample above the Nyquist rate	P2
fN = 12 kHz · x(n) = 13 cos 2π(1/5)n − 5 sin 2π(2/5)n · ya(t) = 13 cos 2000πt − 5 sin 4000πt (aliasing)	P3
At Fs = 40 Hz, f2 = 50 Hz is an alias of f1 = 10 Hz (so are 90 Hz, 130 Hz, ...)	P4
Δ = 0.1, L = 11, |eq(n)| ≤ Δ/2 = 0.05, b = 4 bits	P5
a) N1 = 60, N2 = 2 · b) Ω_o = 0, π/8, π/4, π/2, π → N = ∞, 16, 8, 4, 2 · c) N = 12	P6

SECTION 02

مسائل الامتحانات القديمة

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p. 20	x(t) = cos(200πt), Fs = 400 Hz	P8 ☐
p. 22	x(t) = 10 cos(1500πt), Fs = 3000 samples/s	P9 ☐
p. 24	x(t) = 10 cos(1000πt), Fs = 2000 sample/s	P10 ☐
p. 26	x(t) = A cos(2πf$_o$t), A = 2 V, f$_o$ = 10 kHz	P11 ☐
p. 28	x(t) = 2 cos(2000πt), Fs = 3000 Hz then Fs = 8000 Hz	P12 ☐
p. 30	x(t) = 3 sin(100πt), Fs = 300 samples/s	P13 ☐
p. 33	Is Fs suitable? (Fs = Nyquist rate exactly)	P14 ☐
p. 35	Two signals with two sampling frequencies	P15 ☐
p. 37	A/D then D/A with different sampling periods (Sheet 1.8)	P16 ☐

P7 متكرر ●9×

Periodicity of continuous-time and discrete-time signals

Check the periodicity of the following signals and find their fundamental periods:
(i) $x_a(t) = 3\cos(5t + \pi/6)$ (ii) $x(n) = \cos(3\pi n)$ (iii) $x(n) = \cos(0.01\pi n)$ (iv) $X(n) = 2\sin(3\pi n/105)$ (v) $x(t) = \cos(3000\pi t)$ (vi) $y(t) = 7\cos(400\pi t)$ (vii) $x(n) = \sin(2n + \pi/4)$

(i)

xa(t) = 3 cos(5t + π/6) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 5 \rightarrow f = 5/2\pi \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 2\pi/5 \text{ sec}}$$
$$x_a(t + Tp) = 3\cos(5(t + 2\pi/5) + \pi/6) = 3\cos(5t + 2\pi + \pi/6) = x_a(t)$$

∴ Periodic, fundamental period Tp = 2π/5 sec.

(ii)

x(n) = cos(3πn) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 3\pi \rightarrow F = 3/2 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 2) = \cos(3\pi n + 6\pi) = \cos(3\pi n) = x(n)$$

∴ Periodic, fundamental period **N = 2 samples**.

(iii)

x(n) = cos(0.01πn) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 0.01\pi \rightarrow F = 1/200 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 200) = \cos(0.01\pi n + 2\pi) = x(n)$$

∴ Periodic, fundamental period **N = 200 samples**.

(iv)

x(n) = 2 sin(3πn/105) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 3\pi/105 = \pi/35 \rightarrow F = 1/70 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$

مسائل الامتحانات القديمة

$$x(n + 70) = 2 \sin(\pi n/35 + 2\pi) = x(n)$$

∴ Periodic, fundamental period **N = 70 samples**.

(v)

x(t) = cos(3000πt) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 3000\pi \rightarrow f = 1500 \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 1/1500 \text{ sec} = 0.667 \text{ msec}}$$
$$x(t + Tp) = \cos(3000\pi t + 2\pi) = x(t)$$

∴ Periodic, fundamental period Tp = 1/1500 sec = 0.667 msec.

(vi)

y(t) = 7 cos(400πt) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 400\pi \rightarrow f = 200 \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 5 \text{ msec}}$$
$$y(t + Tp) = 7 \cos(400\pi t + 2\pi) = y(t)$$

∴ Periodic, fundamental period Tp = 5 msec.

(vii)

x(n) = sin(2n + π/4) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 2 \rightarrow F = 2/2\pi = 1/\pi$$

F cannot be expressed as a ratio of two integers → **not periodic**.

النتيجة Tp = 2π/5 s · N = 2 · N = 200 · N = 70 · Tp = 1/1500 s · Tp = 5 ms · sin(2n + π/4) not periodic

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 3** · Sec. 1.5.1 · p. 3
1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 3- Ch- 1.pdf

المحاضرة **LEC 3** · Sec. 1.5.2 · p. 9
1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 3- Ch- 1.pdf

جه في Midterm 26-7-2026 · Midterm 31-7-2023 · Midterm 1 Model-2

مسائل الامتحانات القديمة

Midterm 8-12-2020 Final 27-6-2021 Final 12-1-2019 Final 23-12-2017 Final 2-1-2017 Midterm OLD

5-Exams/Final/ · 5-Exams/Midterm/

مسائل الامتحانات القديمة

P8 متكرر ×8

x(t) = cos(200πt), Fs = 400 Hz

For the signal x(t) = cos(200πt) to be sampled with Fs = 400 Hz:
(i) Find an expression for its corresponding sequence x(n). Then calculate its fundamental period N_P.
(ii) Find the sample values of the first cycle in x(n), then plot x(n) and x(t) on the same figure.

$$x(t) = 1\cos(200\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 1,\quad 2\pi f = 200\pi \rightarrow \mathbf{f = 100\ Hz}$$

$$F_N = 2f_{max} = 2 \times 100 = \mathbf{200\ Hz}$$
$$Fs = 400\ Hz \geq 2f_{max} = 200\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(i)

If the signal is sampled at Fs = 400 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = x_a(nT) = x_a(n/Fs) = 1\cos(200\pi\, n/400) = \mathbf{1\cos(\pi/2\, n)}$$

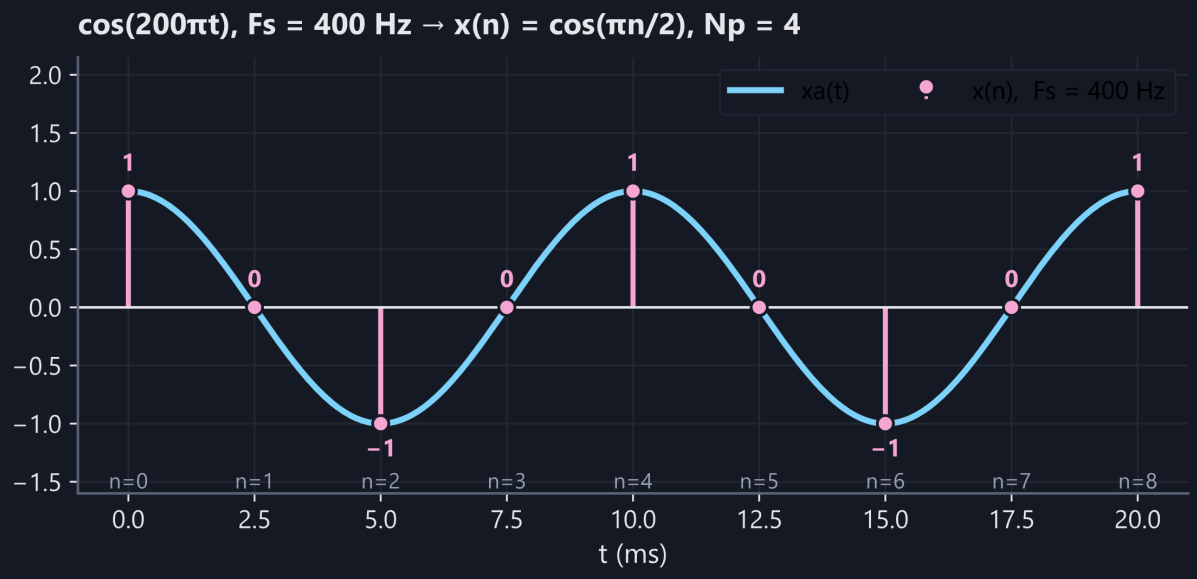
$$F = f/Fs = 100/400 = \mathbf{1/4} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 4\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 3):

$$x(0) = 1\cos(0) = \mathbf{1}$$
$$x(1) = 1\cos(\pi/2) = \mathbf{0}$$
$$x(2) = 1\cos(\pi) = \mathbf{-1}$$
$$x(3) = 1\cos(3\pi/2) = \mathbf{0}$$

(ii) Plot x(t) and x(n) on the same figure:

مسائل الامتحانات القديمة



النتيجة `fN = 200 Hz, no aliasing · x(n) = cos(πn/2), Np = 4: 1, 0, −1, 0`

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 4** · Sec. 1.6.1 · p. 8
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جه في
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`5-Exams/Final/ · 5-Exams/Midterm/ · 6-Extra/`

×3 **P9**

x(t) = 10 cos(1500πt), Fs = 3000 samples/s

For x(t) = 10 cos(1500πt):
(i) If it is sampled with Fs = 3000 sample/s, is it possible to reconstruct this signal from its samples? Find the Nyquist frequency F_N.
(ii) Using Fs, find x(n), then calculate the sample values during one cycle and plot x(t) and x(n).

$$x(t) = 10\cos(1500\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 10,\quad 2\pi f = 1500\pi \rightarrow \mathbf{f = 750\ Hz}$$

(i), (ii)

$$F_N = 2f_{max} = 2 \times 750 = \mathbf{1500\ Hz}$$
$$Fs = 3000\ Hz \geq 2f_{max} = 1500\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(iii), (iv)

If the signal is sampled at Fs = 3000 Hz (t = nT = n/Fs), the discrete-time signal is:

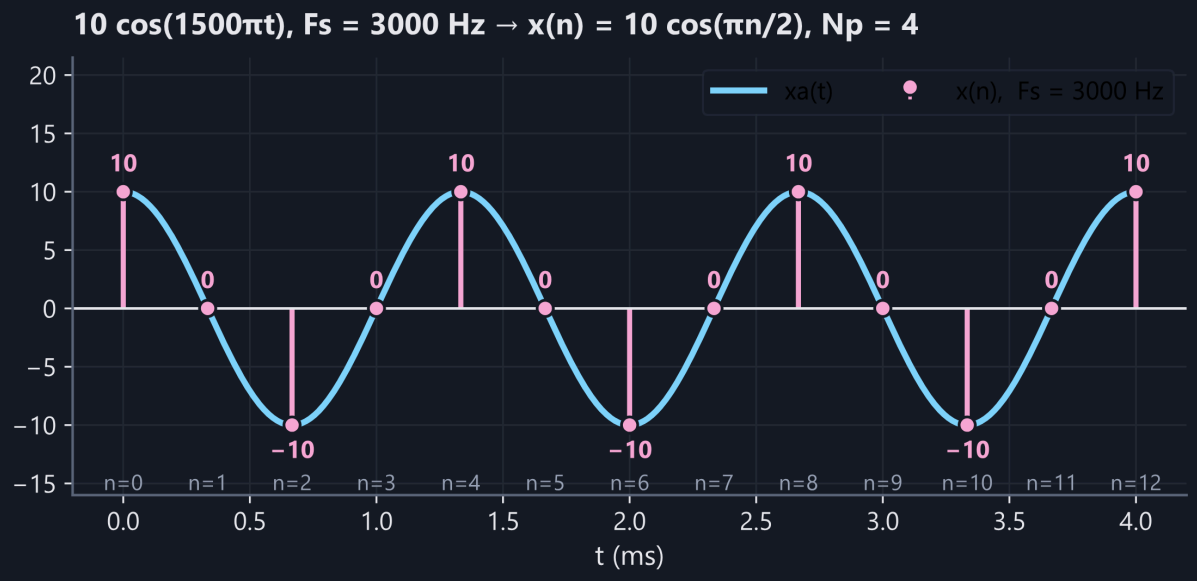
$$x(n) = x_a(nT) = x_a(n/Fs) = 10\cos(1500\pi\, n/3000) = \mathbf{10\cos(\pi/2\, n)}$$

$$F = f/Fs = 750/3000 = \mathbf{1/4} = k/N_p \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_p = 4\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 3):

$$x(0) = 10\cos(0) = \mathbf{10}$$
$$x(1) = 10\cos(\pi/2) = \mathbf{0}$$
$$x(2) = 10\cos(\pi) = \mathbf{-10}$$
$$x(3) = 10\cos(3\pi/2) = \mathbf{0}$$

مسائل الامتحانات القديمة



النتيجة fN = 1500 Hz, no aliasing · x(n) = 10 cos(πn/2), Np = 4: 10, 0, −10, 0

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مسائل الامتحانات القديمة

×3 **P10**

x(t) = 10 cos(1000πt), Fs = 2000 sample/s

For x(t) = 10 cos(1000πt):

(i) If sampled with Fs = 2000 sample/s, is it possible to reconstruct it successfully?

(ii) Find the Nyquist frequency F_N.

(iii) Using F_N, find an expression for x(n).

(iv) Find the values of x(n) in the first cycle.

$$x(t) = 10\cos(1000\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 10,\quad 2\pi f = 1000\pi \rightarrow \mathbf{f = 500\ Hz}$$

(i), (ii)

$$F_N = 2f_{max} = 2 \times 500 = \mathbf{1000\ Hz}$$
$$Fs = 2000\ Hz \geq 2f_{max} = 1000\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(iii) Using the obtained Nyquist frequency F_N = 1000 Hz:

If the signal is sampled at F_N = 1000 Hz (t = nT = n/Fs), the discrete-time signal is:

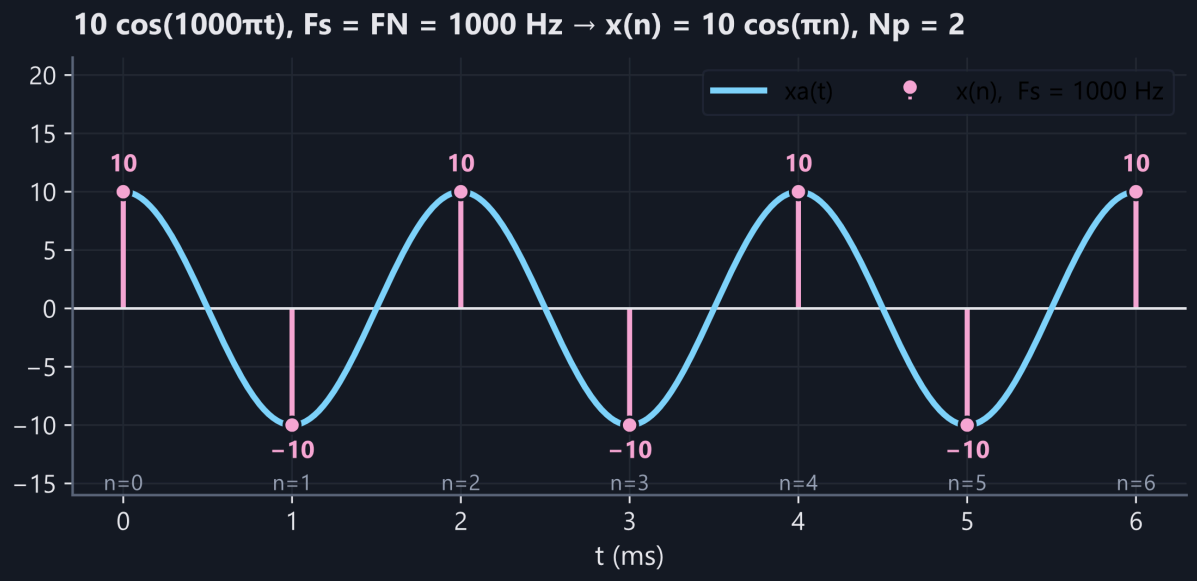
$$x(n) = x_a(nT) = x_a(n/Fs) = 10\cos(1000\pi\, n/1000) = \mathbf{10\cos(\pi\, n)}$$

$$F = f/Fs = 500/1000 = \mathbf{1/2} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 2\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 1):

$$x(0) = 10\cos(0) = \mathbf{10}$$
$$x(1) = 10\cos(\pi) = \mathbf{-10}$$

مسائل الامتحانات القديمة



النتيجة fN = 1000 Hz, no aliasing · at FN: x(n) = 10 cos(πn), Np = 2: 10, −10

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مسائل الامتحانات القديمة

P11 متكرر ●4×

$x(t) = A\cos(2\pi f_o t)$, $A = 2$ V, $f_o = 10$ kHz

For $x_a(t) = A\cos(2\pi f_o t)$, $A = 2$ volt, $f_o = 10$ kHz:

a) Check the periodicity and find the fundamental period.

b) If sampled at Fs = 5 kHz, is it possible to reconstruct the original signal?

c) What is the minimum sampling frequency?

d) Using it, find x(n) and the sample values during one cycle (or the first four samples).

$$x_a(t) = A\cos(2\pi f_0 t),\ A = 2\text{ V},\ f_0 = 10\text{ kHz} \rightarrow x_a(t) = 2\cos(20000\pi t)$$

a) For every fixed value of f, xa(t) is periodic: **$T_p = 1/f_o = 1/10^4 = 0.1$ msec**.

b)

$$F_N = 2f_{max} = 2 \times 10 = \mathbf{20\text{ kHz}}$$
$$Fs = 5\text{ kHz} < 2f_{max} = 20\text{ kHz}$$

Since $Fs < 2f_{max}$, aliasing occurs and the signal **cannot be reconstructed** from its samples.

c) The minimum sampling frequency is **$Fs = 2f_{max} = 20$ kHz**.

d)

If the signal is sampled at Fs = 20000 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = x_a(nT) = x_a(n/Fs) = 2\cos(20000\pi\, n/20000) = \mathbf{2\cos(\pi\, n)}$$

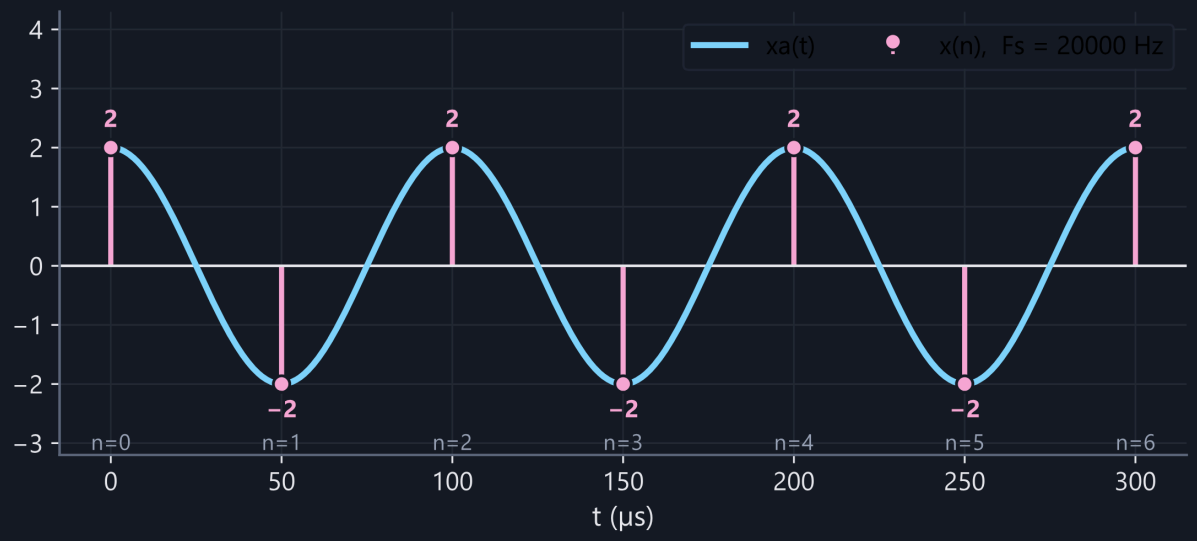
$$F = f/Fs = 10000/20000 = \mathbf{1/2} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 2\text{ samples}}$$

The sample values in the first cycle (n = 0, 1, ..., 1):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi) = \mathbf{-2}$$

First four samples: x(0) = 2, x(1) = −2, x(2) = 2, x(3) = −2

مسائل الامتحانات القديمة



النتيجة Fs = 5 kHz < 20 kHz → aliasing · Fs,min = 20 kHz · x(n) = 2 cos(πn): 2, −2, 2, −2

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المحاضرة **LEC 4** · Sec. 1.6.2 · p. 17
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2-Section/Revision Sheet (summer)/ · 5-Exams/Final/ · 5-Exams/OLD Quiz/

مسائل الامتحانات القديمة

×2 P12

x(t) = 2 cos(2000πt), Fs = 3000 Hz then Fs = 8000 Hz

Given x(t) = 2 cos(2000πt) to be sampled with Fs = 3000 Hz:
(i) Could the original signal be reconstructed successfully from its samples?
(ii) Take Fs = 8000 Hz; find x(n) and its fundamental period N_P.
(iii) Find the sample values of the first cycle and plot x(n).

$$x(t) = 2\cos(2000\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 2,\quad 2\pi f = 2000\pi \rightarrow \mathbf{f = 1000\ Hz}$$

(i)

$$F_N = 2f_{max} = 2 \times 1000 = \mathbf{2000\ Hz}$$
$$Fs = 3000\ Hz \geq 2f_{max} = 2000\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(ii), (iii)

If the signal is sampled at Fs = 8000 Hz (t = nT = n/Fs), the discrete-time signal is:

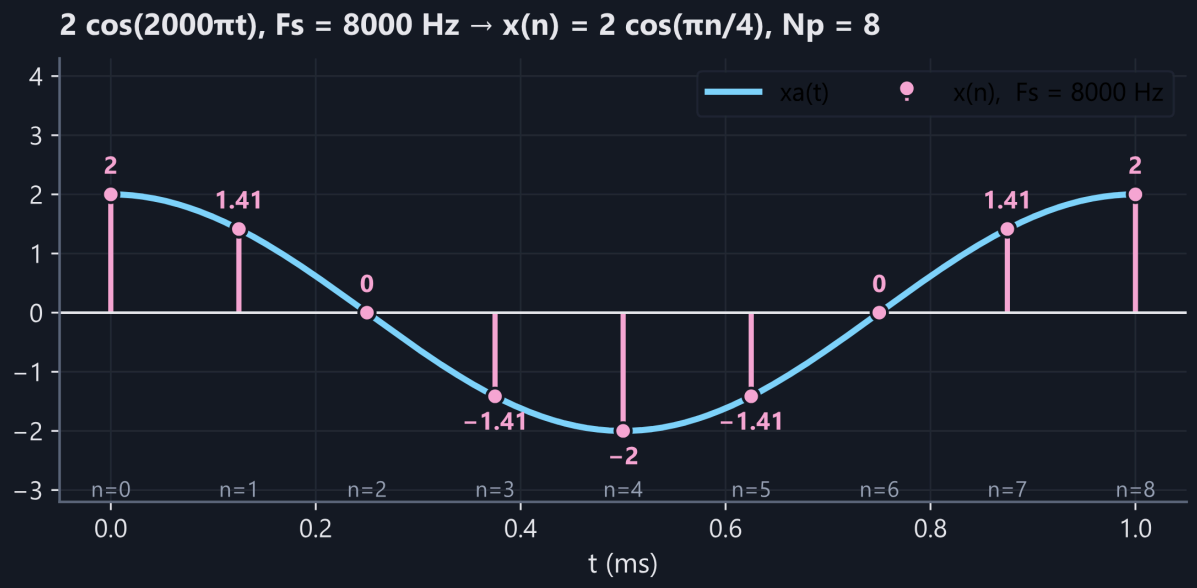
$$x(n) = x_a(nT) = x_a(n/Fs) = 2\cos(2000\pi\, n/8000) = \mathbf{2\cos(\pi/4\, n)}$$

$$F = f/Fs = 1000/8000 = \mathbf{1/8} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 8\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 7):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi/4) = \mathbf{1.414}$$
$$x(2) = 2\cos(\pi/2) = \mathbf{0}$$
$$x(3) = 2\cos(3\pi/4) = \mathbf{-1.414}$$
$$x(4) = 2\cos(\pi) = \mathbf{-2}$$
$$x(5) = 2\cos(5\pi/4) = \mathbf{-1.414}$$
$$x(6) = 2\cos(3\pi/2) = \mathbf{0}$$
$$x(7) = 2\cos(7\pi/4) = \mathbf{1.414}$$

مسائل الامتحانات القديمة



النتيجة No aliasing · Fs = 8 kHz: x(n) = 2 cos(πn/4), Np = 8: 2, 1.414, 0, −1.414, −2, −1.414, 0, 1.414

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5-Exams/Midterm/

مسائل الامتحانات القديمة

×5 متكرر P13

x(t) = 3 sin(100πt), Fs = 300 samples/s

Consider x(t) = 3 sin(100πt):

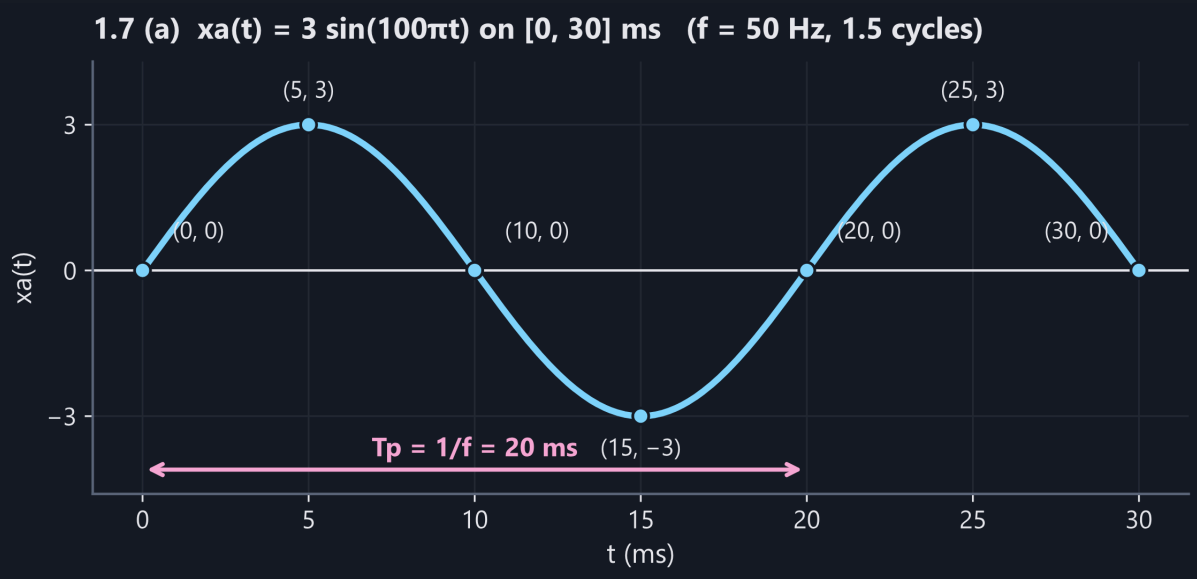
(i) Sketch the signal for [0, 30] ms.

(ii) The signal is sampled at Fs = 300 samples/s. Determine the frequency of x(n) and show that it is periodic.

(iii) Compute the sample values in one period of x(n). Sketch x(n) on the same diagram with x(t). What is the period of x(n)?

(iv) Can you find Fs such that x(n) reaches its peak value 3? What is the minimum Fs?

(i) xa(t) = 3 sin(100πt): A = 3, 2πf = 100π → f = 50 Hz, Tp = 1/f = 20 msec. The interval [0, 30] ms contains 1.5 periods.



(ii), (iii)

If the signal is sampled at Fs = 300 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = x_a(nT) = x_a(n/F_s) = 3\sin(100\pi\, n/300) = \mathbf{3\sin(\pi/3\, n)}$$

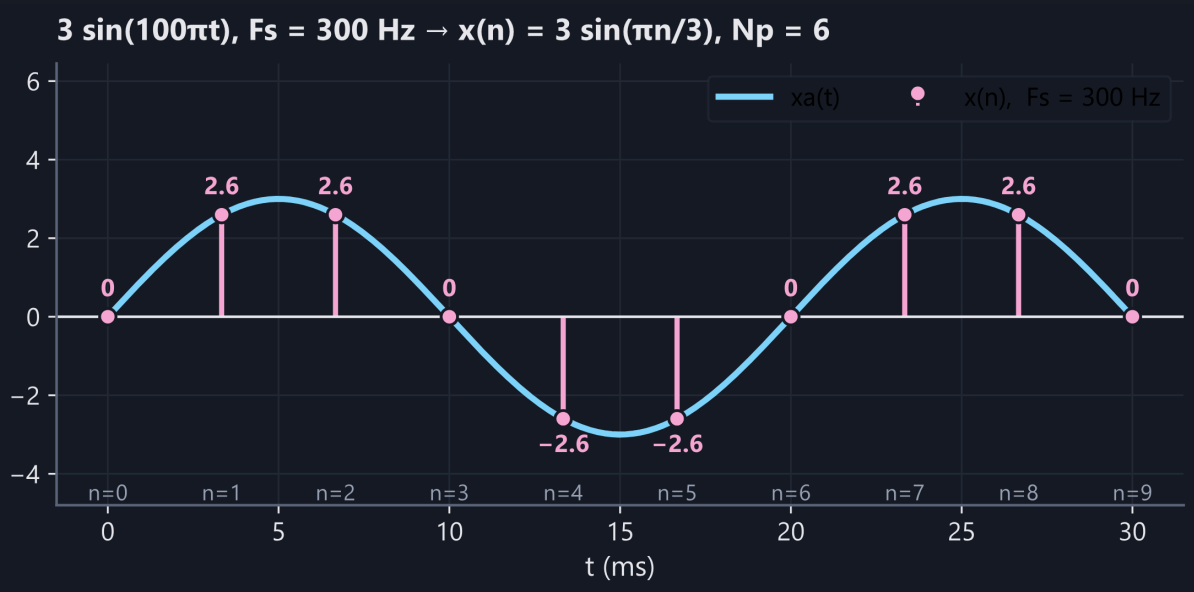
$$F = f/F_s = 50/300 = \mathbf{1/6} = k/N_p \;\rightarrow\; \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_p = 6 \text{ samples}}$$

The sample values in the first cycle (n = 0, 1, ..., 5):

مسائل الامتحانات القديمة

$$x(0) = 3\sin(0) = 0$$
$$x(1) = 3\sin(\pi/3) = 2.598$$
$$x(2) = 3\sin(2\pi/3) = 2.598$$
$$x(3) = 3\sin(\pi) = 0$$
$$x(4) = 3\sin(4\pi/3) = -2.598$$
$$x(5) = 3\sin(5\pi/3) = -2.598$$

Period of x(n): N = 6 samples = 6 × (1/300) s = 20 msec.



(iv) x(n) = 3 sin(100πn/Fs) reaches its peak value 3 when:

$$\sin(100\pi n/Fs) = 1 \rightarrow 100\pi n/Fs = \pi/2 \rightarrow Fs = 200n \rightarrow \text{minimum at } n = 1: \mathbf{Fs = 200 \text{ samples/sec}}$$

النتيجة x(n) = 3 sin(πn/3), N = 6 samples = 20 ms · peak value reached when Fs = 200 samples/s

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مسائل الامتحانات القديمة

Final 4-3-2021 · Q1-b Sheet 1 · 1.7

2-Section/Sheets/ · 5-Exams/Final/ · 5-Exams/Midterm/

مسائل الامتحانات القديمة

×3 **P14**

Is Fs suitable? (Fs = Nyquist rate exactly)

(a) x(t) = 2 cos(3000πt), Fs = 3 kHz. (b) x(t) = 2 cos(2000πt), Fs = 2 kHz. (c) x(t) = cos(300πt), Fs = 300 Hz.

Is Fs suitable? If so, find x(n), the values in the first cycle, and plot.

(a)

$$x(t) = 2\cos(3000\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 2,\quad 2\pi f = 3000\pi \rightarrow \mathbf{f = 1500\ Hz}$$

$$F_N = 2f_{max} = 2 \times 1500 = \mathbf{3000\ Hz}$$
$$Fs = 3000\ Hz \geq 2f_{max} = 3000\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

$Fs = 2f_{max}$ exactly (the Nyquist rate): it is the minimum rate. Sampling at a rate higher than the Nyquist rate is preferred (Example 1.2).

If the signal is sampled at Fs = 3000 Hz (t = nT = n/Fs), the discrete-time signal is:

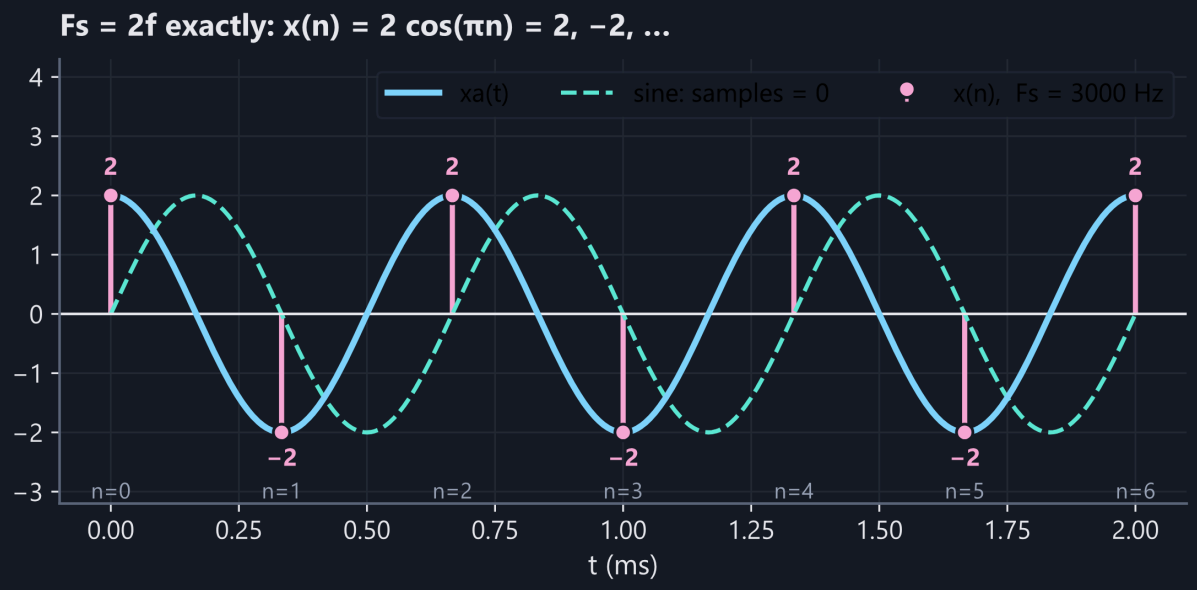
$$x(n) = x_a(nT) = x_a(n/Fs) = 2\cos(3000\pi\, n/3000) = \mathbf{2\cos(\pi\, n)}$$

$$F = f/Fs = 1500/3000 = \mathbf{1/2} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 2\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 1):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi) = \mathbf{-2}$$

مسائل الامتحانات القديمة



(b) x(t) = 2 cos(2000πt), Fs = 2 kHz → f = 1000 Hz, Fs = 2f → x(n) = 2 cos(πn): 2, −2.

(c) x(t) = cos(300πt), Fs = 300 Hz → f = 150 Hz, Fs = 2f → x(n) = cos(πn): 1, −1.

النتيجة

```
Fs = 2fmax exactly → x(n) = 2 cos(πn): 2, −2 (works, but a rate
above Nyquist is preferred)
```

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5-Exams/Final/ · 7-Books (References)/Extract Question From Book/

×1 **P15**

Two signals with two sampling frequencies

x1(t) = 5 cos 200πt and x2(t) = 2 sin 500πt are sampled with Fs1 = 400 Hz and Fs2 = 1 kHz respectively.
(i) Are the sampling frequencies suitable? (ii) If suitable, find x1(n) and x2(n).

x1(t):

$$x1(t) = 5\cos(200\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 5,\quad 2\pi f = 200\pi \rightarrow \mathbf{f = 100\ Hz}$$

$$F_N = 2f_{max} = 2 \times 100 = \mathbf{200\ Hz}$$
$$Fs = 400\ Hz \geq 2f_{max} = 200\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

If the signal is sampled at Fs = 400 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = xa(nT) = xa(n/Fs) = 5\cos(200\pi\, n/400) = \mathbf{5\cos(\pi/2\ n)}$$

$$F = f/Fs = 100/400 = \mathbf{1/4} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 4\ samples}$$

x2(t):

$$x2(t) = 2\sin(500\pi t) \rightarrow 2\pi f = 500\pi \rightarrow \mathbf{f = 250\ Hz}$$

$$F_N = 2f_{max} = 2 \times 250 = \mathbf{500\ Hz}$$
$$Fs = 1000\ Hz \geq 2f_{max} = 500\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

If the signal is sampled at Fs = 1000 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = xa(nT) = xa(n/Fs) = 2\sin(500\pi\, n/1000) = \mathbf{2\sin(\pi/2\ n)}$$

$$F = f/Fs = 250/1000 = \mathbf{1/4} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 4\ samples}$$

∴ Both sampling frequencies are suitable.

مسائل الامتحانات القديمة

النتيجة x1: Np = 4 · x2: Np = 4 · both sampling frequencies are suitable (no aliasing)

Key words x1(t) x2(t)

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Book Midterm 2015 Model-B
7-Books (References)/Extract Question From Book/

مسائل الامتحانات القديمة

×2 **P16**

A/D then D/A with different sampling periods (Sheet 1.8)

A signal processing system: $x_a(t)$ → A/D → $x(n)$ → D/A → $y_a(t)$. The sampling periods are T1 = 5 ms (A/D) and T2 = 1 ms (D/A). Determine $y_a(t)$ if $x_a(t)$ = 3 cos(100πt) + 2 sin(250πt).

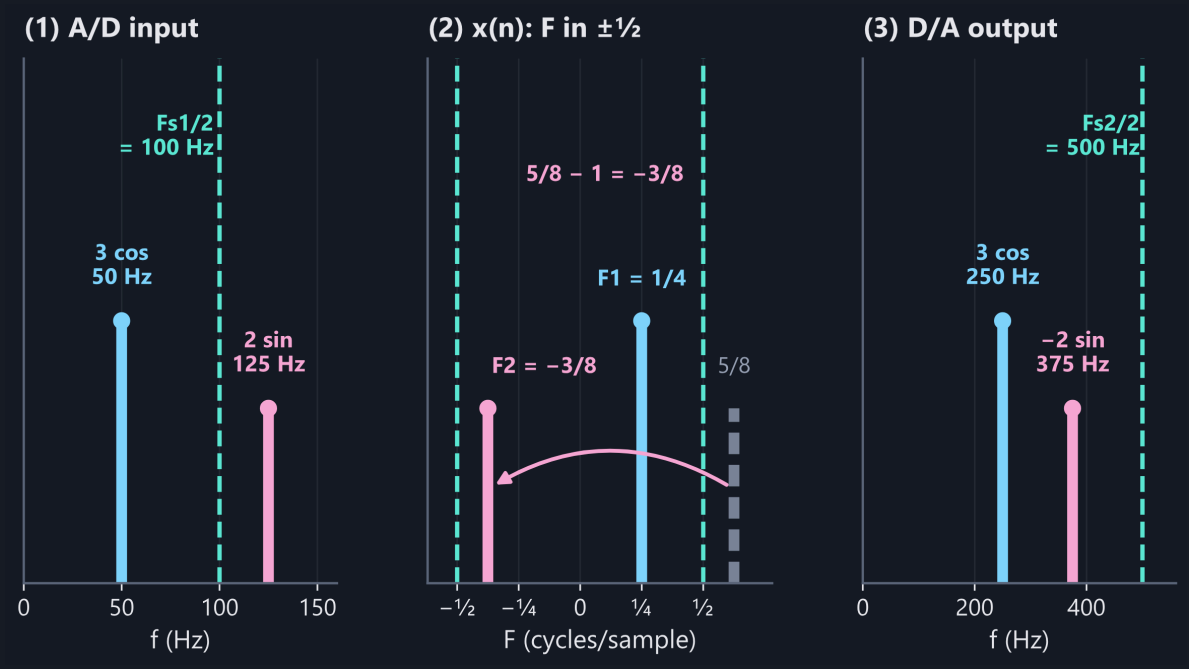
f1 = 50 Hz, f2 = 125 Hz. A/D: Fs = 1/T1 = 200 Hz → folding frequency 100 Hz.

$$x(n) = 3\cos 2\pi(1/4)n + 2\sin 2\pi(5/8)n = 3\cos 2\pi(1/4)n + 2\sin 2\pi(1 - 3/8)n$$

$$\mathbf{x(n) = 3\cos(\pi n/2) - 2\sin(3\pi n/4)}$$

D/A: T2 = 1 ms → n = t/T2 = 1000t:

$$\mathbf{y_a(t) = 3\cos(500\pi t) - 2\sin(750\pi t)}$$



النتيجة $x(n) = 3\cos(\pi n/2) - 2\sin(3\pi n/4)$ · $y_a(t) = 3\cos(500\pi t) - 2\sin(750\pi t)$

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 4** · Example 1.3 · p. 44

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مسائل الامتحانات القديمة

جه في

Revision Sheet 1 · Q12 | Sheet 1 · 1.8

2-Section/Revision Sheet (summer)/ · 2-Section/Sheets/

★ Cheat sheet · مسائل الامتحانات القديمة

النتايج النهائية

Tp = 2π/5 s · N = 2 · N = 200 · N = 70 · Tp = 1/1500 s · Tp = 5 ms · sin(2n + π/4) not periodic	P7
fN = 200 Hz, no aliasing · x(n) = cos(πn/2), Np = 4: 1, 0, −1, 0	P8
fN = 1500 Hz, no aliasing · x(n) = 10 cos(πn/2), Np = 4: 10, 0, −10, 0	P9
fN = 1000 Hz, no aliasing · at FN: x(n) = 10 cos(πn), Np = 2: 10, −10	P10
Fs = 5 kHz < 20 kHz → aliasing · Fs,min = 20 kHz · x(n) = 2 cos(πn): 2, −2, 2, −2	P11
No aliasing · Fs = 8 kHz: x(n) = 2 cos(πn/4), Np = 8: 2, 1.414, 0, −1.414, −2, −1.414, 0, 1.414	P12
x(n) = 3 sin(πn/3), N = 6 samples = 20 ms · peak value reached when Fs = 200 samples/s	P13
Fs = 2fmax exactly → x(n) = 2 cos(πn): 2, −2 (works, but a rate above Nyquist is preferred)	P14
x1: Np = 4 · x2: Np = 4 · both sampling frequencies are suitable (no aliasing)	P15
x(n) = 3 cos(πn/2) − 2 sin(3πn/4) · ya(t) = 3 cos(500πt) − 2 sin(750πt)	P16

SECTION 03

Sheet 1

الجزء النظري

الأسئلة

p. 40 Define: Signal, System, Signal processing (state one example for each). 1.1 ☐

p. 42 Classify the different types of signal processing. 1.2 ☐

p. 44 Classify the different signal types. 1.3 ☐

p. 46 Explain the basic three operations performed in A/D converters. 1.4 ☐

p. 47 State and explain the sampling theorem. 1.5 ☐

1.1

Define: Signal, System, Signal processing (state one example for each).

1.1 Define the following items: Signal, system, signal processing. (State one example for each).

- **Signal:** is defined as any physical quantity that varies with time, space, or any other independent variable(s). Mathematically, we describe a signal as a function of one or more independent variables.
  Examples: $s1(t) = 5t$ (varies linearly with t), $s2(t) = 20t^2$ (varies quadratically with t), and $s(x, y) = 3x + 2xy + 10y^2$ (two independent variables, e.g. the spatial coordinates in a plane). A speech signal cannot be described by such an expression; a segment of it is represented as a sum of several sinusoids of different amplitudes, frequencies and phases.
- **System:** is defined as a physical device that performs an operation on a signal.
  Example: a filter used to reduce the noise and interference corrupting a desired information. If the operation is linear, the system is called linear; if it is nonlinear, the system is nonlinear.
- **Signal processing:** when we pass a signal through a system, as in filtering, we say that we have processed the signal.
  Example: filtering the noise and interference from the desired signal. Digital processing may be performed by digital hardware (logic circuits), by software, or by a combination of both.

الفكرة

Signal = varying physical quantity; system = device that operates on it; processing = passing the signal through the system.

Key words: Signal | System | Signal processing

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المحاضرة

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Sheet 1

Sheet 1 · Q 1.1 الشيت

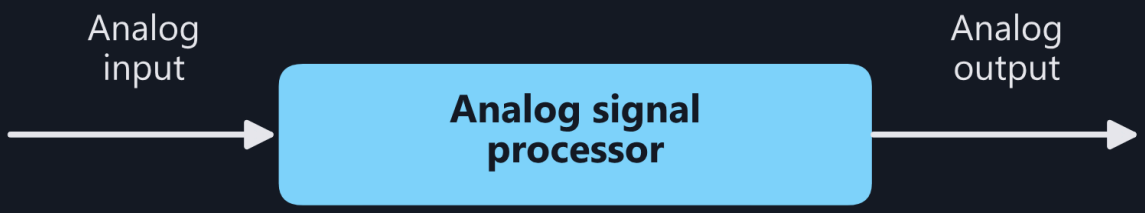
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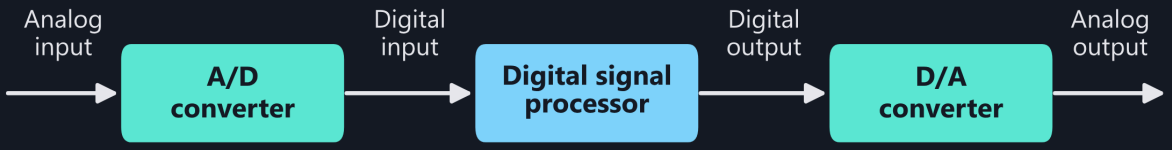
1.2

Classify the different types of signal processing.

There are two types of signal processing systems (Sec. 1.2):

1. **Analog signal processor (Fig. 1.1):** the analog input signal is processed directly by an analog signal processor to give an analog output signal.
2. **Digital signal processor (Fig. 1.2):** provides an alternative method for processing the analog signal. The **A/D converter** converts the analog input into a digital signal (a sequence of numbers), the **digital signal processor** performs the required operations on it, and the **D/A converter** converts the result back into an analog output signal.





الفكرة Analog signal processing (direct) and digital signal processing (A/D → DSP → D/A).

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المحاضرة **LEC 2** · Sec. 1.2 · p. 11

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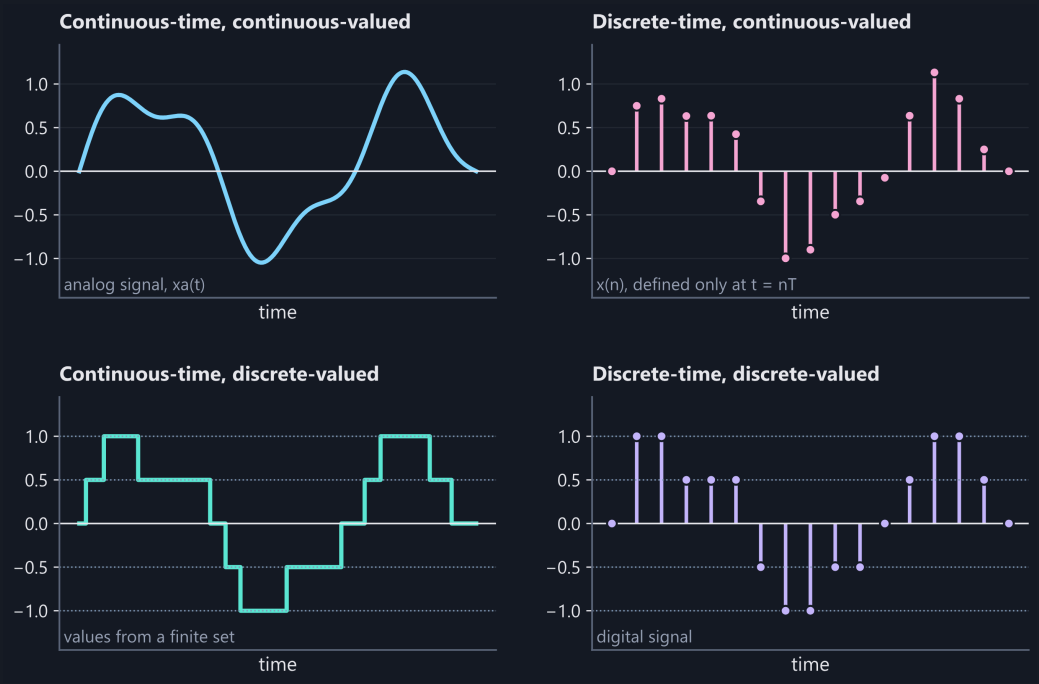
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1.3

Classify the different signal types.

1. **Continuous-time vs. discrete-time:** continuous-time (analog) signals are defined for every value of time and take on values in the continuous interval (a, b). Discrete-time signals are defined only at certain specific values of time.
2. **Continuous-valued vs. discrete-valued:** a continuous-valued signal takes on all possible values on a finite or an infinite range. A discrete-valued signal takes on values from a finite set of possible values.
3. **Deterministic vs. random:** a deterministic signal can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule. Otherwise, the signal is random.
4. **Multichannel vs. multidimensional:** a multichannel signal is generated by multiple sources or sensors and represented in vector form, $S3(t) = [s1(t)\ s2(t)\ s3(t)]^T$. A signal is M-dimensional if its value is a function of M independent variables.



Continuous/discrete time, continuous/discrete valued, deterministic/random, multichannel/multidimensional.

الفكرة

Sheet 1

Key words Deterministic vs. random

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المحاضرة

LEC 2 · Sec. 1.4 · p. 18

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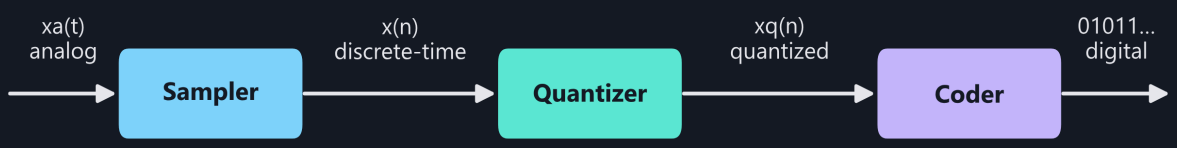
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1.4

Explain the basic three operations performed in A/D converters.

To process analog signals by digital means, it is first necessary to convert them into a sequence of numbers having finite precision. This is A/D conversion, a three-step process (Fig. 1.9):



1. **Sampling:** the conversion of a continuous-time signal into a discrete-time signal obtained by taking "samples" of the continuous-time signal at discrete-time instants. If xa(t) is the input to the sampler, the output is **x(n) = xa(nT)**, where T is the sampling interval.
2. **Quantization:** the conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal. The value of each sample is selected from a finite set of possible values. The difference $x_q(n) - x(n)$ is the quantization error.
3. **Coding:** each discrete value $x_q(n)$ is represented by a **b-bit binary sequence**.

Sampling does not result in a loss of information if the signal bandwidth is finite (**invertible**), while quantization is a **noninvertible** process that results in signal distortion.

الفكرة
Sampling → quantization → coding.

Key words Sampling Quantization Coding

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المحاضرة
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1.5

State and explain the sampling theorem.

- To select Fs we must have information about the frequency content of the signal, e.g. f_{max} = 3000 Hz for speech and 5 MHz for television signals. The signal may be passed through a filter that severely attenuates components above f_{max}.
- To avoid aliasing, Fs is selected so that **$Fs > 2f_{max}$** (1.47). Then every component $|f_i| < f_{max}$ is mapped into $-1/2 \le F_i = f_i/Fs \le 1/2$ (1.48).

Sampling theorem: if the highest frequency contained in an analog signal xa(t) is $f_{max} = B$ and the signal is sampled at a rate $Fs > 2f_{max} = 2B$, then xa(t) can be exactly recovered from its sample values using the interpolation function:

$$g(t) = \frac{\sin(2\pi Bt)}{2\pi Bt} \quad (1.50) \qquad x_a(t) = \sum x_a(n/Fs)\, g(t - n/Fs) \quad (1.51)$$

- The sampling rate **$f_N = 2B = 2f_{max}$** is called the **Nyquist rate**.
- Because of the complexity and the infinite number of samples required, these reconstruction formulas are primarily of theoretical interest.

الفكرة — If Fs > 2 fmax the analog signal can be recovered exactly from its samples (Nyquist rate = 2 fmax).

Key words: Sampling theorem

المصدر — من الـ PDFs بتاعتك

LEC 4 · Sec. 1.6.2 · p. 17 — المحاضرة
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Sheet 1

Cheat sheet · Sheet 1 ⋆

افتكر

Signal = varying physical quantity; system = device that operates on it; processing = passing the signal through the system.	1.1
Analog signal processing (direct) and digital signal processing (A/D → DSP → D/A).	1.2
Continuous/discrete time, continuous/discrete valued, deterministic/random, multichannel/multidimensional.	1.3
Sampling → quantization → coding.	1.4
If Fs > 2 fmax the analog signal can be recovered exactly from its samples (Nyquist rate = 2 fmax).	1.5

المعادلات

$$g(t) = \sin(2\pi Bt) / (2\pi Bt) \quad (1.50)$$

$$xa(t) = \Sigma\, xa(n/Fs)\, g(t - n/Fs) \quad (1.51)$$

SECTION 04

Sheet 1

المسائل

الأسئلة

p. 50	Which sinusoids are periodic? Compute their fundamental period.	1.6 ☐
p. 52	xa(t) = 3 sin(100πt): sketch, sample at Fs = 300, period, peak value	1.7 ☐
p. 55	A/D with T1 = 5 ms then D/A with T2 = 1 ms: find ya(t)	1.8 ☐

1.6

Which sinusoids are periodic? Compute their fundamental period.

1.6 Determine which of the following sinusoid signals are periodic, and then compute their fundamental period.

(a) xa(t) = 3 cos(5t + π/6) (b) xa(t) = 3 cos(100πt) (c) x(n) = cos[π(30n/105)] (d) x(n) = 2 exp[j(n/6 − π)]

(a)

xa(t) = 3 cos(5t + π/6) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 5 \rightarrow f = 5/2\pi \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 2\pi/5 \text{ sec}}$$
$$x_a(t + T_p) = 3\cos(5(t + 2\pi/5) + \pi/6) = 3\cos(5t + 2\pi + \pi/6) = x_a(t)$$

∴ Periodic, fundamental period Tp = 2π/5 sec.

(b)

xa(t) = 3 cos(100πt) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 100\pi \rightarrow f = 50 \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 20 \text{ msec}}$$
$$x_a(t + T_p) = 3\cos(100\pi(t + 20\text{ ms})) = 3\cos(100\pi t + 2\pi) = x_a(t)$$

∴ Periodic, fundamental period Tp = 20 msec.

(c)

x(n) = cos[π(30n/105)] is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 30\pi/105 \rightarrow F = 1/7 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 7) = \cos(30\pi(n + 7)/105) = \cos(30\pi n/105 + 2\pi) = x(n)$$

∴ Periodic, fundamental period **N = 7 samples**.

(d) x(n) = 2 exp[j(n/6 − π)] = $2e^{-j\pi}\, e^{j2\pi Fn}$: a discrete-time complex exponential is periodic only if its relative frequency is rational.

$$2\pi F = 1/6 \rightarrow F = 1/(12\pi)$$

F cannot be expressed as a ratio of two integers → **not periodic**.

النتيجة	Tp = 2π/5 s · Tp = 20 ms · N = 7 samples · 2 exp[j(n/6 − π)] not periodic

المصدر — من الـ PDFs بتاعتك

المحاضرة — **LEC 3** · Sec. 1.5.1 · p. 3
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المحاضرة — **LEC 3** · Sec. 1.5.2 · p. 9
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الشيت — **Sheet 1** · Q 1.6
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1.7

xa(t) = 3 sin(100πt): sketch, sample at Fs = 300, period, peak value

1.7 Consider the analog sinusoid signal xa(t) = 3 sin(100πt).

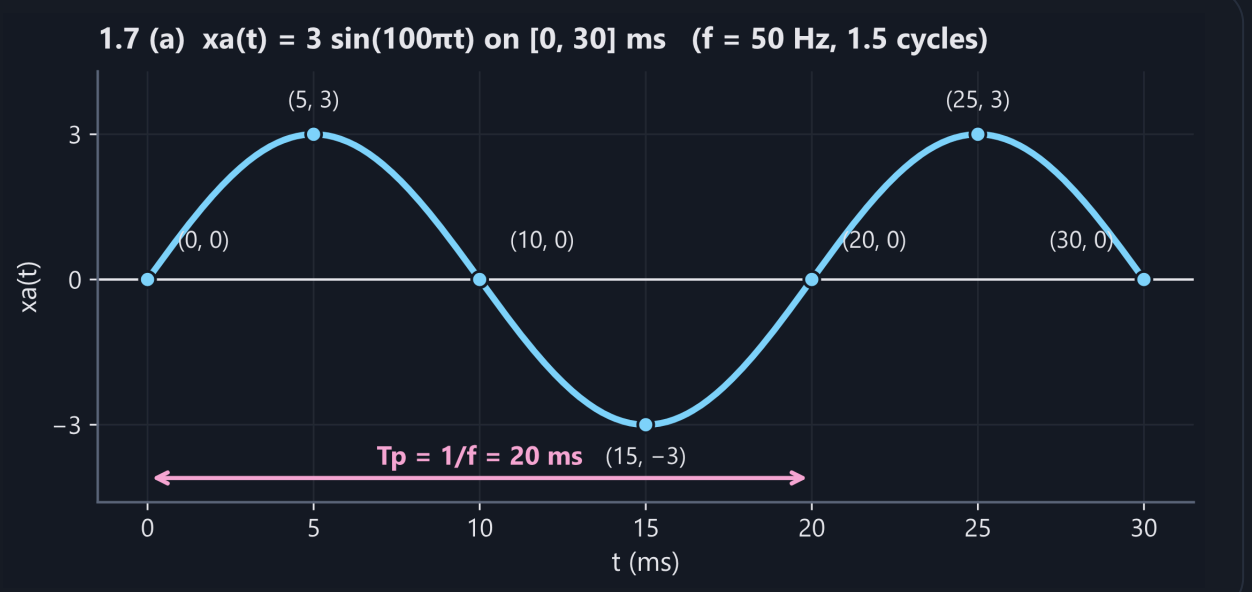
(a) Sketch the signal for [0, 30] ms.

(b) The signal is sampled with sampling rate Fs = 300 samples per second. Determine the frequency of the discrete-time signal and show that it is periodic.

(c) Compute the sample values in one period of x(n). Sketch x(n) on the same diagram with xa(t). What is the period of this discrete-time signal x(n)?

(d) Can you find a sampling rate Fs such that the signal x(n) reaches its peak value 3? What is the minimum Fs suitable for this task?

(a) xa(t) = 3 sin(100πt) = A sin(2πft): A = 3, 2πf = 100π → f = 50 Hz, Tp = 1/f = 20 msec. The interval [0, 30] ms contains 1.5 periods.



(b)

If the signal is sampled at Fs = 300 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = x_a(nT) = x_a(n/F_s) = 3\sin(100\pi\, n/300) = \mathbf{3\sin(\pi/3\, n)}$$

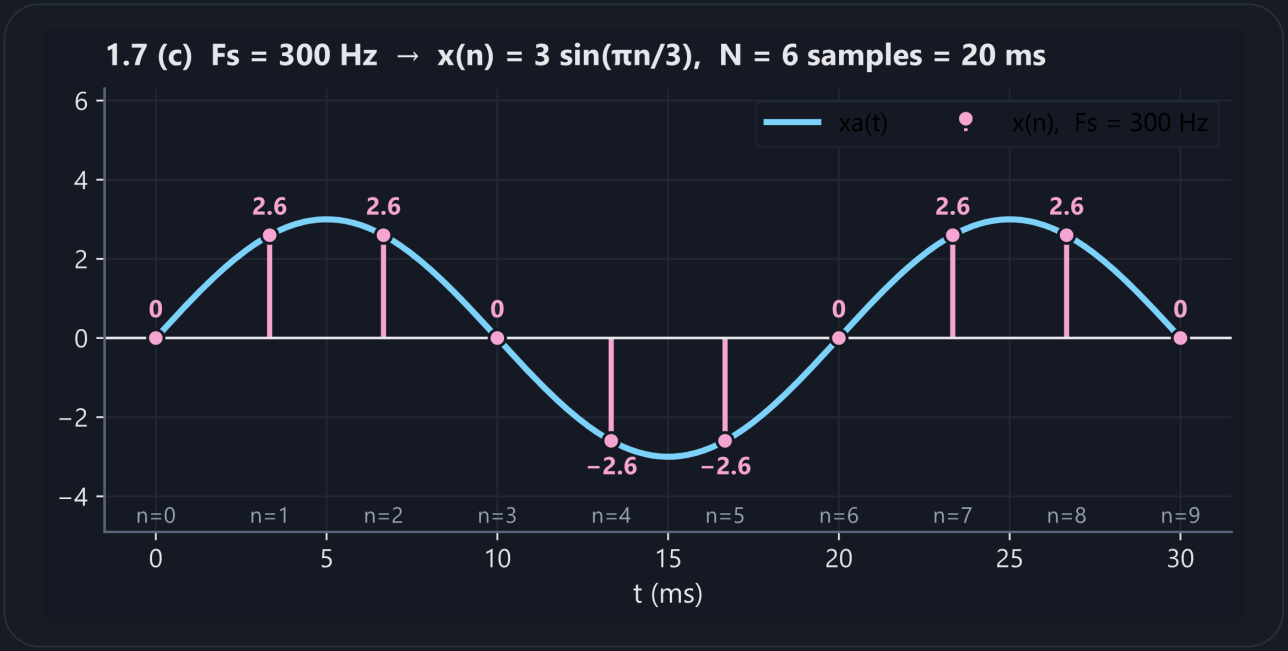
$$F = f/F_s = 50/300 = \mathbf{1/6} = k/N_p \;\rightarrow\; \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_p = 6 \text{ samples}}$$

$$x(n+6) = 3\sin(\pi(n+6)/3) = 3\sin(\pi n/3 + 2\pi) = x(n)$$

(c)

$$x(0) = 3\sin(0) = 0$$
$$x(1) = 3\sin(\pi/3) = 2.598$$
$$x(2) = 3\sin(2\pi/3) = 2.598$$
$$x(3) = 3\sin(\pi) = 0$$
$$x(4) = 3\sin(4\pi/3) = -2.598$$
$$x(5) = 3\sin(5\pi/3) = -2.598$$

The period of x(n) is **N = 6 samples** = 6 × (1/300) s = 20 msec.

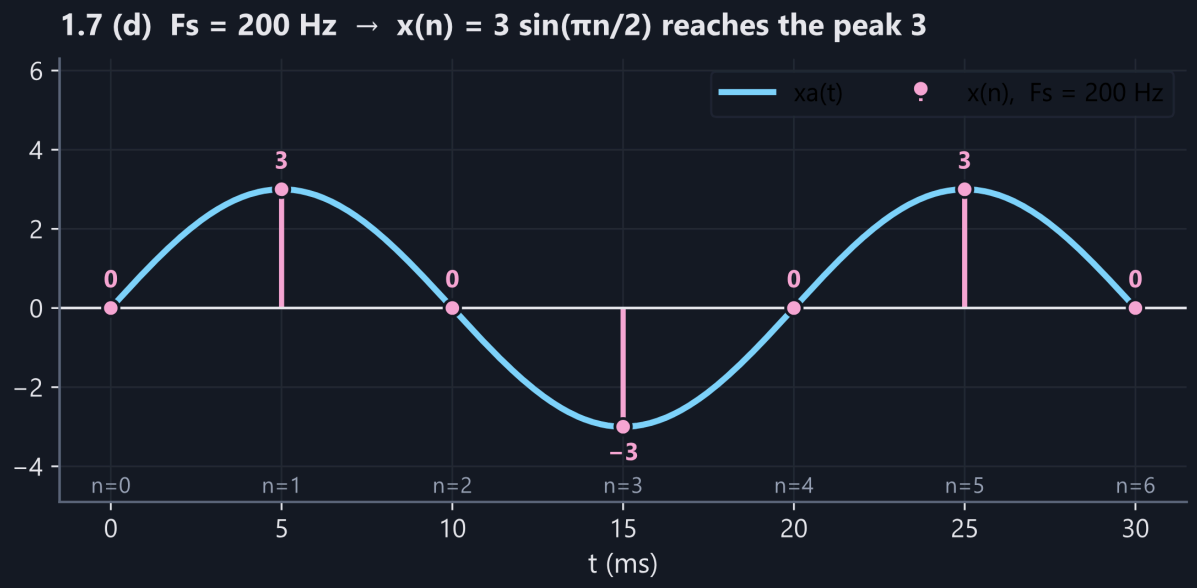


(d) x(n) = 3 sin(100πn/Fs) reaches its peak value 3 when:

$$\sin(100\pi n/Fs) = 1 \rightarrow 100\pi n/Fs = \pi/2 \rightarrow Fs = 200n$$

Yes. The minimum is at n = 1: **Fs = 200 samples/sec**. Then x(n) = 3 sin(πn/2) = 0, 3, 0, −3.

Sheet 1



النتيجة `x(n) = 3 sin(πn/3): 0, 2.598, 2.598, 0, −2.598, −2.598 · N = 6 (20 ms) · Fs = 200 → 0, 3, 0, −3`

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المحاضرة

LEC 4 · Sec. 1.6.1 · p. 8

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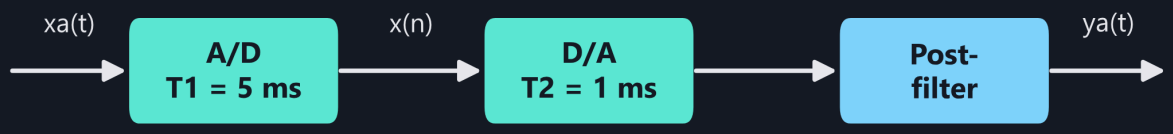
Sheet 1 · Q 1.7

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1.8

A/D with T1 = 5 ms then D/A with T2 = 1 ms: find ya(t)

1.8 Consider a simple signal processing system as shown in the following figure. The sampling periods of the A/D and D/A converters are T1 = 5 ms and T2 = 1 ms respectively. Determine the output ya(t) if the input is given by: xa(t) = 3 cos(100πt) + 2 sin(250πt) (t in second).



The frequencies existing in the analog signal are f1 = 50 Hz (2πf1 = 100π) and f2 = 125 Hz (2πf2 = 250π).

The A/D converter has T1 = 5 ms, so Fs = 1/T1 = 200 samples/s and the folding frequency is Fs/2 = 100 Hz, the maximum frequency that can be represented uniquely by the sampled signal. By making use of (1.30) we obtain:

$$
\begin{aligned}
x(n) &= x_a(nT1) = x_a(n/Fs) \\
&= 3\cos 2\pi(50/200)n + 2\sin 2\pi(125/200)n \\
&= 3\cos 2\pi(1/4)n + 2\sin 2\pi(5/8)n \\
&= 3\cos 2\pi(1/4)n + 2\sin 2\pi(1 - 3/8)n \\
&= 3\cos 2\pi(1/4)n + 2\sin 2\pi(-3/8)n
\end{aligned}
$$

Finally, we obtain:

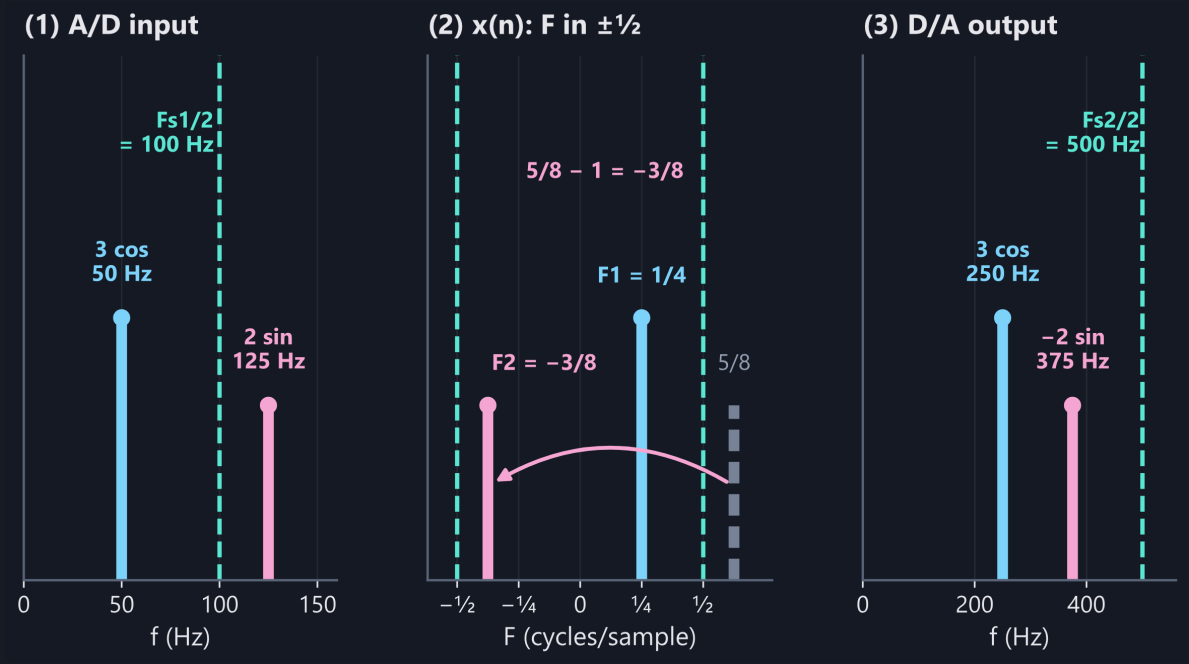
$$\mathbf{x(n) = 3\cos(\pi n/2) - 2\sin(3\pi n/4)}$$

The D/A converter has T2 = 1 ms (Fs = 1000 samples/s). Using t = nT2, i.e. n = t/T2 = 1000t:

$$y_a(t) = 3\cos(\pi(1000t)/2) - 2\sin(3\pi(1000t)/4) = \mathbf{3\cos(500\pi t) - 2\sin(750\pi t)}$$

which is different from xa(t): the distortion is caused by aliasing (Fs = 200 Hz < 2 × 125 Hz), and the frequencies are scaled by T1/T2 = 5.

Sheet 1



النتيجة $x(n) = 3\cos(\pi n/2) - 2\sin(3\pi n/4)$ · $y_a(t) = 3\cos(500\pi t) - 2\sin(750\pi t)$ (aliasing)

بناءً على الـ PDFs المصدر — من الـ

المحاضرة — ***LEC 4*** · *Example 1.3* · *p. 44*
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الشيت — ***Sheet 1*** · *Q 1.8*
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Cheat sheet · Sheet 1 ★

النتايج النهائية

$T_p = 2\pi/5$ s · $T_p = 20$ ms · $N = 7$ samples · $2\exp[j(n/6 - \pi)]$ not periodic	1.6
$x(n) = 3\sin(\pi n/3)$: 0, 2.598, 2.598, 0, −2.598, −2.598 · $N = 6$ (20 ms) · $F_s = 200$ → 0, 3, 0, −3	1.7
$x(n) = 3\cos(\pi n/2) - 2\sin(3\pi n/4)$ · $y_a(t) = 3\cos(500\pi t) - 2\sin(750\pi t)$ (aliasing)	1.8